PRASHANT SURESH KAMBLE

Aerospace Research Engineer | CFD & Scientific Machine Learning

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RESEARCH SUMMARY

Aerospace Research Engineer specializing in reacting flow computational fluid dynamics (CFD), 100% pure hydrogen swirl combustion, Physics-Informed Neural Networks (DA-PINN), and non-linear hyperelastic structural mechanics. Brings rigorous expertise in multi-physics solver development, ASME V&V 20 grid verification, and scientific machine learning, supported by 1 peer-reviewed journal publication, 3 first-author manuscripts under review, and open-access archived research datasets (Figshare).

EDUCATION

Lovely Professional University — *Bachelor of Technology in Aerospace Engineering (First Division | CGPA: 7.69/10)* **Aug 2021 – May 2025**

PEER-REVIEWED PUBLICATIONS & PREPRINTS

Flame Bifurcation, Flashback Margins, and NOx Scaling in a Swirl-Stabilized Hydrogen Aero-Engine Combustor **2026**

DOI: 10.1515/ot-2026-0033 | Figshare DOI | GitHub Code

Authors: **P. S. Kamble** *(Sole Author)* | *Journal: Open Transport (De Gruyter Brill, In Press, DOI: 10.1515/ot-2026-0033)*

- Pioneered 3D reacting CFD simulations across flight throttles (Φ = 0.55–1.00) incorporating curvature-corrected SST k-ω turbulence, dual-rate finite-rate/eddy-dissipation kinetics, and 72-direction Discrete Ordinates radiation under ASME V&V 20 grid standards (GCI_{21} = 0.040%).
- Discovered fuel-momentum-driven topological flame transition from an attached M-flame at idle to an aerodynamically lifted V-flame at cruise/takeoff (Φ ≥ 0.895).
- Established boundary-layer flashback immunity (AFMI ≥ 3.42 > 1.0) and mapped NO_x kinetics transition from N_2O-intermediate dominance at idle to thermal Zeldovich dominance at takeoff (n = 4.92 scaling).

Physics-Informed Neural Networks for 3D Aerodynamic and Reacting Flow Reconstruction in a Hydrogen Swirl Aero-Engine Combustor (DA-PINN) **2026**

SSRN DOI | Figshare DOI | GitHub Code

Authors: **P. S. Kamble** *(Sole Author)* | *Journal: Physics of Fluids (AIP Publishing, Under Review) / SSRN Electronic Journal*

- Engineered a 4D Fourier-ResNet DA-PINN surrogate enforcing continuous 3D Favre-RANS momentum, continuity, swirl balance, thermal energy, and species transport directly via autograd loss functions.
- Formulated dynamic ReLoBRaLo loss balancing and a two-stage optimization protocol (AdamW + L-BFGS), eliminating spectral bias and gradient starvation across stiff reaction fronts.
- Demonstrated zero-shot holdout accuracy (R^2 = 0.9523 for W_z, 0.9479 for T), delivering a >4,500× inference speedup (1.58 s full 3D volume) and >7,000× model compression (14.8 GB → 2.1 MB).

Functionalized Carbon Nanotube–Natural Rubber Composites for Aircraft Landing Gear Shock Absorber Disks **2026**

SSRN DOI | Figshare DOI | GitHub Code

Authors: **P. S. Kamble** *(First Author), M. Dhawan, et al.* | *Journal: Journal of Applied Physics (AIP Publishing, Under Review) / SSRN Electronic Journal*

- Synthesized silane-grafted f-MWCNT/NR composites (0–5.0 phr); identified a 3.0 phr mechanical percolation threshold (+58.9% tensile strength, +320% hysteretic loss dissipation).
- Calibrated 3rd-order Yeoh hyperelastic strain energy functions across concurrent dual-mode tension (600%) and compression (50%), ensuring Drucker stability.
- Conducted 3D ANSYS Explicit Dynamics drop impact simulations under FAA FAR Part 23.725 (200 kg mass, 3.05 m/s sink rate), proving full airworthiness compliance (3.46 G ≤ 3.50 G limit).

Project Setu: 3D Multi-Physics Design and Scaled Structural Analysis for a Relativistic Lightsail Architecture **2026**

arXiv:2609.06156 | Figshare DOI | GitHub Code

Authors: **P. S. Kamble** *(Sole Author)* | *Journal: Acta Astronautica (Elsevier, Under Review)*

- Constructed a 3D multi-physics framework coupling Ansys Lumerical 3D Maxwell FDTD wave optics (99.7821% reflectance, <10 ppm absorption), Stefan-Boltzmann thermal radiation (923.02 K steady-state), and ANSYS SHELL181 non-linear prestressed modal FEA under 100 GW laser irradiance (0.20c target velocity).
- Cross-verified radiation thrust formulation against JAXA IKAROS (0.12% error) and NASA LightSail 2 (2.13% error) spaceflight telemetry.

PROFESSIONAL R&D EXPERIENCE

AeroMyne — *CFD Engineer – Combustion & Propulsion (Solapur, India)* **Jan 2025 – Present**

- Architected high-fidelity 3D reacting CFD pipelines in ANSYS Fluent for hydrogen swirl aero-combustors, integrating non-adiabatic thermal transport and finite-rate chemistry to match laser diagnostic benchmarks within <0.5% peak temperature error.
- Implemented curvature-corrected SST k-ω turbulence closures & Discrete Ordinates (DO) radiation, capturing complex precessing vortex cores (PVC), vortex breakdown dynamics, and thermal shear layer mixing.
- Applied ASME V&V 20 Grid Convergence Index (GCI) verification protocols across production meshes (GCI_{21} = 0.040%), quantifying numerical discretization uncertainty to guarantee defensible models.

Jet Aerospace — *Aircraft Design Intern (Palakkad, India)* **Jun 2024 – Dec 2024**

- Optimized conceptual UAV wing and fuselage geometries in CATIA V5 and SolidWorks, improving aerodynamic lift-to-drag (L/D) ratios by 8% across 5 flight envelope configurations.
- Performed 15+ CFD (ANSYS Fluent) and FEA (ANSYS Mechanical) simulation cycles, reducing structural stress concentration points by 12% under peak gust loads.
- Attained a 5% airframe structural weight reduction by embedding electronic hardware layouts into load-bearing composite panels.

SELECTED COMPUTATIONAL R&D PROJECTS

NASA ARN2 Compressible Nozzle CFD & Acoustic Turbulence | GitHub Code — 2025

- Modeled subsonic jet development (M_{jet} = 0.51, Re_D = 5,601) from NASA Glenn Acoustic Reference Nozzle 2 using Fluent density-based coupled solvers.
- Validated SST k-ω turbulence against NASA PIV data (12.06% MAPE), establishing grid independence across structured hex meshes.
- Quantified near-field acoustic shear layer turbulence intensity, achieving 0.73% mean velocity convergence between coarse and fine grids.

Turbulent Pipe Flow & Viscous Sublayer Resolution | GitHub Code — 2025

- Resolved $y^+ \approx 1$ viscous sublayer gradients (y = 0.318 mm) across Re = 10,000 pipe flows in ANSYS Fluent utilizing ICEM CFD hex blocking.
- Correlated friction factors and velocity profiles within 3–5% of Haaland analytical correlations and empirical turbulent pipe benchmarks.
- Assessed near-wall turbulence modeling ($y^+ \approx 1$ vs $y^+ \approx 10$), demonstrating that resolving the laminar sublayer directly is required for wall shear stress fidelity.

Saturn V F-1 Nozzle Bolted Joint FEA | GitHub Code — 2024

- Analyzed 3D non-linear contact & bolt pretension (1,000 lbf) on Inconel 718 flange bolts in ANSYS Mechanical under 4.72 MPa internal pressure and 500°C thermal gradients.
- Evaluated contact pressure distributions and bolt stress relaxation across operational ascent trajectories, certifying a +15% structural safety margin.
- Investigated non-linear gasket and bolt load-transfer analysis, securing complete zero-leakage sealing against hot gas penetration.

CERTIFICATIONS

STK Certification Grandmaster (ANSYS / AGI) — Credential — 2024

Certified SOLIDWORKS Professional (CSWP) — Credential — 2024

Certified 3DEXPERIENCE / CATIA Mechanical Designer – Associate — Credential — 2024

IBM Data Science Professional Certificate — Credential — 2022

TECHNICAL SKILLS & LANGUAGES

Programming & SciAI:	Python, PyTorch, C++, MATLAB
Simulation Suite:	ANSYS Fluent (CFD), ANSYS Mechanical (FEA), Ansys Lumerical (FDTD), NASA CEA
CAD & Mesh Generation:	CATIA V5, SolidWorks, SpaceClaim, ICEM CFD
Physical Domains & Standards:	Reacting Flow & Hydrogen Combustion, Turbulence Modeling (SST k-ω), Physics-Informed Neural Networks (PINNs), ASME V&V 20
Languages & Proficiency:	English (Full Professional Proficiency), Hindi (Native), Marathi (Native)

ACADEMIC & RESEARCH REFERENCES

Avinash Jha — Director of Engineering
AeroMyne Labs | **director@aeromyne.com**
CFD Supervisor

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B.Tech Thesis Supervisor